Aras Ozdemir

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EDUCATION

Cornell University, David A. Duffield College of Engineering, Ithaca, NY
Master of Engineering in Mechanical Engineering, **GPA: 3.93** **Expected May 2026**

Pennsylvania State University, College of Engineering, State College, PA
Bachelor of Science in Biomedical Engineering
Bachelor of Science in Mechanical Engineering, **GPA: 3.55** **May 2024**

Selected Coursework: Systems Engineering & Six Sigma · Innovation and Design of Biomedical Technologies · Healthcare Innovations · Dimensional Tolerancing (GD&T) · Advanced Product Design, Entrepreneurship for Scientists & Engineers

SPECIALIZED SKILLS

Technical: MATLAB, R, Python, COMSOL Multiphysics, SolidWorks, ANSYS, Excel, LaTeX, GD&T
Quality Systems:-CAPA, FMEA/DFMEA, DHF, Design Control, NCR, IQ/OQ/PQ Validation, ISO 13485, ISO 9001, 510(k) Submissions
Management: Design Control Processes, Risk Management, Decision-Making, Data Analysis, English *(Fluent)*, German *(Fluent)*
Certifications: Lean Six Sigma Black Belt

ENGINEERING EXPERIENCE

Engineering Lead, Capstone Project*, NY* **August 2025 - Present**

- Lead An ASML sponsored project, managing the integration of an industrial robotic handling arm.
- Design and implement electrical and mechanical support systems to enable safe laboratory deployment and operation.
- Perform dynamic force simulations in MATLAB to evaluate loading conditions and structural safety margins.
- Conduct FMEA-based risk assessments to identify operational hazards and define mitigation strategies.
- Architect a sensor-integrated fail-safe shutdown system to immediately disable the robot under unsafe trajectory.

Quality Engineer, Fresenius Kabi, *Frankfurt, Germany* **November 2024 – August 2025**

- Supported preparation of **Class II 510(k)** submission documentation, contributing to successful regulatory filing.
- Investigated and resolved reported device complaints, performing root cause analysis to support corrective actions.
- Revised 40+ technical drawings **for FDA 21 CFR Part 820** and **ASME Y14.5** compliance, standardizing **GD&T tolerances.**
- Engineered an automated magnetic hatch durability tester **(10,000 cycles),** reducing **~28 manual labor** hours per validation run and improving repeatability by minimizing operator variability.
- Reengineered validation test workflows to eliminate steps and enable material reuse**, improving efficiency by ~20%**

Design Engineer, B. Braun medical inc., *State College, PA* **January 2024 – May 2024**

- Analyzed failure modes of a secondary connector unclamping during procedures using mechanical testing.
- Designed and iterated CAD prototypes supported by COMSOL-based FEA simulations to optimize stress distribution
- **Improved connector grip strength by 48%** while maintaining customer-specified flow rate requirements.
- Conducted validation testing and ensured documentation complied with FDA design control and regulatory standards.

Research Assistant, The Pennsylvania State University, *State College, PA* **January 2023 – January 2024**

- Conducted extensive research on antibiotic properties of Bacteriophages.
- Utilized advanced microscopy techniques to conduct detailed and comprehensive examinations of the interaction between bacteriophage and bacteria through high-resolution photography.
- Developed a MATLAB application with a graphical user interface, streamlining data analysis through automation.
- Implemented multiple advanced filtering methods with precision and expertise to effectively minimize data noise, resulting in the generation of visually clear and informative graphs that enhanced data presentation and interpretation.